Surga: a fitted closure for the production-to-dissipation ratio across seven classes of wall-bounded flow

Sidharth SS
Independent Researcher

21 October 2026

Abstract

This note documents a compact expression for the ratio of turbulent kinetic energy production to dissipation, P/ε, intended to be used across the whole thickness of a wall-bounded layer and not only in its inner part. The expression was obtained during a first application exercise of a new symbolic-regression algorithm, and was then interpreted physically, refitted and tested by hand. It is a function of four quantities that a RANS solver already has at every node — wall distance in wall units, wall distance normalised by the layer thickness, the local shear stress and the stress at the edge of the layer — and of nine fitted constants. On a 52-profile benchmark spanning channels, pipes, plane Couette flow, zero- and adverse-pressure-gradient boundary layers, an aerofoil wake and a compressible channel, the fit gives a pooled R^2 of 0.970 (RMSE 0.072), a mean leave-one-profile-out score of 0.958 and a mean leave-one-class-out score of 0.941. For comparison, the two closures printed in a recent study of the same quantity — an analytic form and a symbolic-regression form, both functions of y^+ alone — score 0.483 and 0.488 on the same points over the whole layer, and 0.958 and 0.958 on the inner domain they were designed for, where the present fit gives 0.973. We read the difference as a consequence of what the two forms can and cannot represent outside the inner layer, and we say so with the caveats it deserves: this is an empirical exercise, not a derivation, and the constants are fitted.

1 Scope

This is a working note rather than a paper. It records one expression, the data it was fitted to, and the tests we ran on it, in enough detail that someone else can reproduce or refute it. It arose as a test application of a symbolic-regression algorithm under development; the structure reported in Section 4 is the interpretable form that survived that exercise, after which it was re-fitted by conventional least squares and checked on held-out cases. The physics entered in choosing which variables to allow and in reading the resulting constants, not as a derivation of the final formula.

All comparisons are made against the two closures printed in Jiang & Rosic [1]: their Eq. (27), referred to throughout as *M16*, and their Eq. (35), referred to as *PySR*. Both are transcribed exactly as printed and evaluated with the printed constants, with no refitting of any kind, on exactly the same points as the present closure.

2 The quantity and three constraints on it

The turbulent kinetic energy budget of a stationary parallel shear flow is

$$P - \varepsilon + T + P_p + D_\nu = 0, \tag{1}$$

Table 1: Measured near-wall behaviour, $\log P/\varepsilon$ against $\log y^+$ over $0.5 < y^+ < 5$.

flow class	measured exponent	coefficient ($\times 10^{-3}$)	points
plane channel	3.06	5.32	125
pipe	3.06	5.00	114
plane Couette	3.09	5.07	27
ZPG boundary layer	3.08	5.25	76
APG boundary layer	3.12	6.76	74
aerofoil wake	3.07	7.63	22
compressible channel	3.03	3.63	125

Table 2: Position and height of the buffer-layer maximum of P/ε.

flow class	peak at y^+	peak P/ε
plane channel	11.8	1.78
pipe	11.7	1.79
plane Couette	11.4	1.76
ZPG boundary layer	11.9	1.76
APG boundary layer	11.1	1.74
aerofoil wake	11.2	1.72
compressible channel	13.1	1.91

with $P = -\langle u'v' \rangle \, \mathrm{d}U/\mathrm{d}y$ the production, ε the dissipation, T the turbulent transport and P_p the pressure-transport contribution. Rearranged,

$$\frac{P}{\varepsilon} - 1 = -\frac{T + P_p}{\varepsilon}. \tag{2}$$

Equation (2) is the reason the ratio is worth modelling and the reason it is hard: the departure from unity is exactly the net transport. A closure for P/ε is therefore a closure for transport, and it cannot be right by accident.

Three properties of P/ε are known well enough to constrain any closure.

The wall. No-slip kinematics give $\langle u'v' \rangle^+ \propto (y^+)^3$ and hence $P/\varepsilon = c_w (y^+)^3 + O((y^+)^4)$ as $y^+ \to 0$. The exponent is not adjustable. Fitting $\log P/\varepsilon$ against $\log y^+$ over $0.5 < y^+ < 5$ in the present benchmark gives the exponents in Table 1: they are 3.03–3.12 in every class. The coefficient is 5.0–5.3×10^{-3} in the incompressible canonical flows, which reproduces independently the channel-DNS value of 4.96×10^{-3} quoted in [1].

The buffer peak. Production switches on before dissipation responds, so the ratio overshoots. In this benchmark the maximum is found at almost the same place and reaches almost the same height in every class (Table 2), although the friction Reynolds number there spans two decades and the pressure gradient, geometry and Mach number all change.

The outer layer. Beyond the log layer the ratio has no universal value, shape or limit. In this benchmark it sits near 1.15 in a zero-pressure-gradient boundary layer, near 1.5–1.6 under an adverse pressure gradient, near 0.65 at the symmetry plane of a plane Couette flow, and near zero at the centreline of a channel. This is the region where closures differ, and it is the region this note is mostly about.

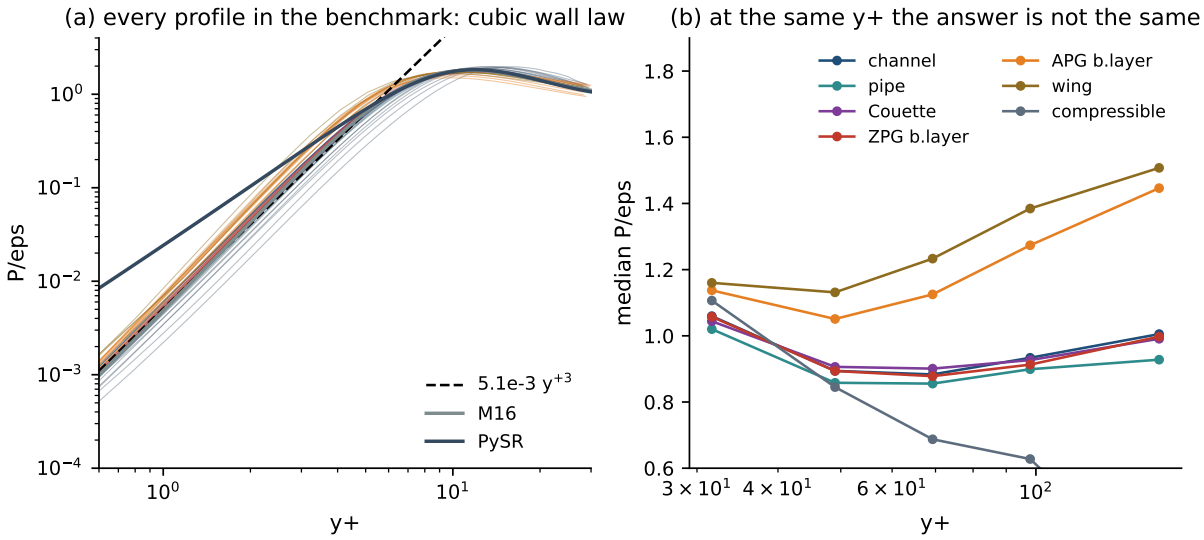


Figure 1: (a) All 52 profiles for $y^+ < 30$ against the cubic wall law $5.1 \times 10^{-3}(y^+)^3$. The inner layer collapses; the two published forms are also shown. (b) The same data in the outer layer, as a median per class in bins of y^+. At the same y^+ the classes give different answers, and the ordering follows how much stress the outer flow carries rather than the Reynolds number.

3 Why a function of y^+ alone is limited here

Both published closures are functions of y^+ alone. Two measurements suggest that this is what limits them outside the inner layer.

First, at fixed y^+ the answer depends on the flow. Taking the median of each profile over $y^+ \in [60, 120]$, the observed P/ε has a 10th–90th percentile range of 0.71–1.36 about a median of 0.91; over $y^+ \in [30, 60]$ the range is 0.87–1.15. A single-valued function of y^+ must map all 52 profiles onto one number at a given y^+.

Second, within a single profile the ratio is not monotone in y^+: it rises from zero at the wall to about 1.8 at $y^+ \approx 12$, settles back to about 1 in the log layer, and then either rises again under an adverse pressure gradient or collapses at a free-stream edge. A monotone function cannot do all three.

What is missing is the local stress. We therefore carry, in addition, the normalised total stress $\tau(\eta)$ and define the *stress surplus*

$$X(\eta) \;=\; \max\{\,\tau(\eta) - [\,1 + (\tau_e - 1)\eta\,],\; 0\,\}, \tag{3}$$

where τ is normalised by the wall value and $\tau_e = \tau(1)$ is its value at the edge of the layer. The bracket is the stress the wall itself would have supplied, decaying linearly between the wall and the edge; in a channel it decays exactly linearly and $X \equiv 0$. Any excess over it arrived from outside — from a pressure gradient, from a moving wall, from the wake of an aerofoil.

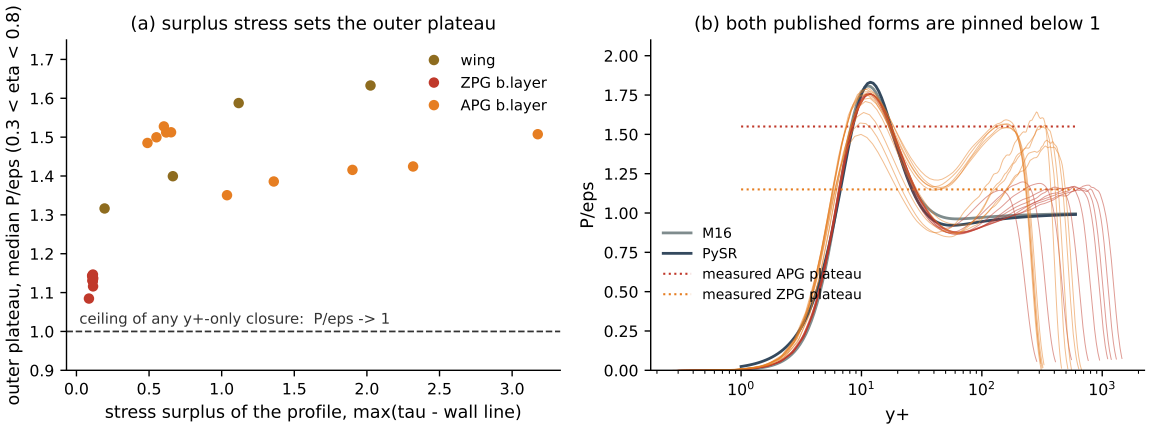


Figure 2: (a) Outer plateau of P/ε against the maximum stress surplus of the profile, for the 23 free-stream flows in the benchmark. Zero-pressure-gradient layers cluster near 1.1–1.15, adverse pressure gradient and aerofoil cases near 1.35–1.63, and the rise appears to saturate. (b) The two published closures against the same data. Their large-y^+ limit is $1/(1 + a_4/y^+)$, which approaches unity from below.

Table 3: Constants of Eqs. (5)–(8).

symbol	value	interpretation
a_1	0.17331	not fitted: cube root of the measured near-wall coefficient 5.2×10^{-3}
p_w	3	fixed by the no-slip kinematic constraint
b_1	0.22516	decay of the buffer bump; its peak sits near $y^* \approx 2/b_1$
c_1	0.10237	amplitude of the buffer bump
k	3.80596	strength of the outer cut-off
r_{conf}	5.01887	sharpness of the cut-off at a wall or symmetry plane
r_{free}	8.05979	sharpness of the cut-off at a free-stream edge
a	0.51082	ceiling of the surplus factor, $1 + a = 1.51$
q	3.73557	how far out in the layer the surplus acts
g	78.65106	gain of the stress-surplus term
h	1.89912	surplus a free edge produces with no pressure gradient

4 The closure

The expression is a product of three factors, one for each of the three regimes above:

$$y^* = y^+\sqrt{\tau}, \tag{4}$$

$$W = \tanh^{p_w}(a_1 y^*)\Big[1 - T^5(1 - T^2) + c_1 y^{*2} e^{-b_1 y^*}\Big], \quad T = \tanh(b_1^2 y^*/2), \tag{5}$$

$$D = \exp\big[-k\,\eta^r\,(1 - \tau_e)\big], \qquad r = \begin{cases} r_{\mathrm{free}} & \text{free-stream edge} \\ r_{\mathrm{conf}} & \text{wall or symmetry plane} \end{cases} \tag{6}$$

$$B = 1 + a\,\tanh\big[\eta^q(gX + h(1 - \tau_e))\big], \tag{7}$$

$$\frac{P}{\varepsilon} = W \cdot D \cdot B. \tag{8}$$

Here η is the wall distance normalised by the distance to the point where the shear stress vanishes rather than by δ_{99}, and X is the surplus of Eq. (3). The constants are listed in Table 3.

4.1 The wall factor

The cube is the kinematic constraint of Section 2 and the exponent is held at $p_w = 3$. The constant a_1 is not fitted: it is the cube root of the measured near-wall coefficient, $a_1 = (5.2 \times 10^{-3})^{1/3} =$

0.17331, so that $W \to 5.2 \times 10^{-3}(y^+)^3$ as the wall is approached. The bracket carries the buffer peak: the term $c_1 y^{*2} e^{-b_1 y^*}$ peaks near $y^* \approx 2/b_1 \approx 9$, close to the measured peak position of Table 2, and in practice does the work of the production overshoot. The small $T^5(1 - T^2)$ term sharpens the peak slightly instead of letting it broaden.

The wall distance inside W is $y^* = y^+\sqrt{\tau}$, not y^+. This is a local, stress-based inner scale: in an adverse pressure gradient the stress at $y^+ = 100$ can be several times its wall value, so the same y^+ is not the same physical place. At the wall $\tau = 1$ and $y^* = y^+$, so the asymptotic behaviour of Eq. (5) is unaffected.

4.2 The edge factor

A layer ends either because its stress has been used up or because the geometry closes it. The factor $(1 - \tau_e)$ measures the first: it is the fraction of the wall stress lost by the edge — about 0.99 in a channel, a pipe or a boundary layer, and zero in a Couette flow, where the stress is constant all the way to the moving wall. D therefore switches itself off in Couette flow, which is right about the stress and, as Section 9 explains, is where the largest error of the present fit comes from.

The exponent r controls how abruptly the layer ends. We allow two values, one for a free-stream edge and one for a wall or a symmetry plane; the fit prefers a sharper cut-off at a free stream ($r_{\text{free}} = 8.06$) than at a closed boundary ($r_{\text{conf}} = 5.02$). Two values were needed; one could not describe both, with the zero-pressure-gradient class losing about 0.09 in R^2 when the two are merged.

4.3 The surplus factor

Where $X = 0$ — in a channel, identically — the surplus factor is unity and the model reduces to the wall function multiplied by the edge factor. Where the outer flow carries surplus stress the ratio rises, and the rise appears to saturate: the measured plateau moves from 1.13 at zero pressure gradient to 1.63 at the strongest adverse gradient and then stops rising however large the surplus becomes (Fig. 2a). The tanh in Eq. (7) is that statement, with a ceiling of $1 + a = 1.51$.

The term $h(1 - \tau_e)$ is the part of the surplus a free-stream edge produces with no pressure gradient at all. It is what lifts the zero-pressure-gradient boundary layers to about 1.15 while leaving a channel — whose edge is a symmetry plane rather than a free stream and whose X is zero — at about unity. The weight η^q with $q \approx 3.7$ confines the surplus term to the outer part of the layer.

5 Fitting and validation

The four inputs are available at every point of every profile, so the fit is a nine-dimensional nonlinear least squares over 8,751 points. The residual minimised is the plain difference between model and data — the same quantity the reported R^2 is built from — so no part of the profile is privileged by the objective. Eight random starts were used and the best kept. Fit and tests are in `validate6.py`; the closure itself is `surga5.py`.

Two tests matter more than the in-sample score. In the first, leave-one-profile-out, the nine constants are re-fitted from scratch to 51 profiles and applied, unchanged, to the 52nd; this is repeated for all 52. The mean score is 0.958, the weakest profile is 0.810 (Couette flow at $Re_\tau = 500$), and no profile falls below 0.5. In the second, leave-one-class-out, a whole flow class is removed and the constants re-fitted on the remaining six before the missing class is scored.

Table 4: In-sample and held-out scores by class (pooled over all points of the class).

flow class	profiles	in-sample R^2	whole class held out
plane channel	13	0.989	0.972
pipe	4	0.983	0.977
plane Couette	3	0.912	0.912
ZPG boundary layer	9	0.986	0.956
APG boundary layer	10	0.918	0.894
aerofoil wake	4	0.914	0.909
compressible channel	9	0.966	0.965
all / mean	52	0.970	0.941

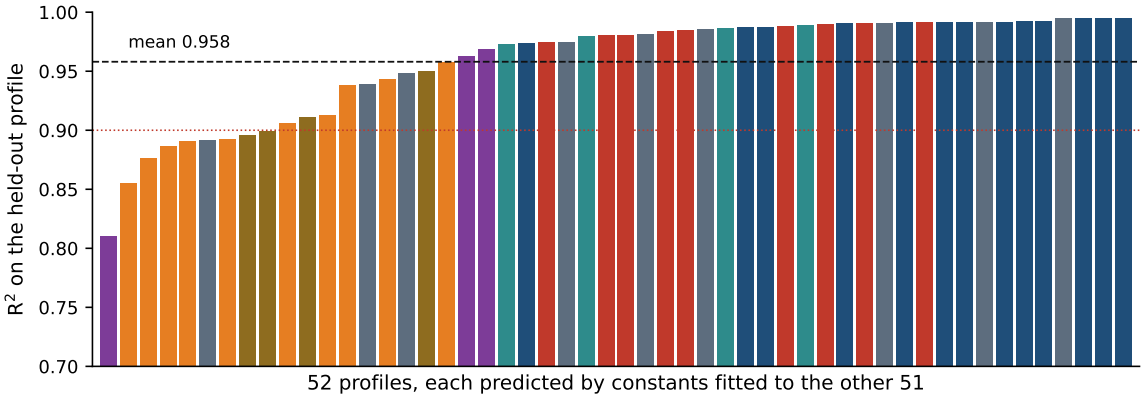


Figure 3: Leave-one-profile-out scores for all 52 profiles, ordered. Each profile is predicted by constants fitted to the other 51.

6 Results across the seven classes

Table 5 collects the pooled scores over the whole layer, and Fig. 4 the class-median profiles. The closure reproduces the buffer peak and the outer behaviour of all seven classes; the residuals are largest where the layer carries surplus stress.

7 The two published closures

The comparison is with the two forms printed in [1], both a ratio of hyperbolic tangents:

$$\text{M16 (Eq. 27):} \quad \frac{P}{\varepsilon} = \frac{\tanh^2(a_1 y^+)}{\tanh(a_2 y^+ - a_3) + a_4/y^+}, \tag{9}$$

$$\text{PySR (Eq. 35):} \quad \frac{P}{\varepsilon} = \frac{\tanh(a_1 y^+)}{\tanh(a_2 y^+ - a_3) + a_4/y^+}, \tag{10}$$

with $(a_1, a_2, a_3, a_4) = (0.111,\ 0.052,\ 0.443,\ 2.888)$ for Eq. (9) and $(0.128,\ 0.053,\ 0.620,\ 5.78)$ for Eq. (10). In both, the a_4/y^+ term is in the denominator, so the large-y^+ limit is $1/(1 + a_4/y^+)$, approached from below. The authors report a training-pool global R^2 of 0.987 for M16 and $R^2 \geq 0.978$ on 19 external zero-pressure-gradient profiles for PySR, and state clearly that Eq. (10) is validated only for $1 < y^+ \leq \min(150,\ 0.3Re_\tau)$ and that evaluating it in the wake is an uncontrolled extrapolation. That statement governs how the comparison should be made.

Inside their domain. Restricted to the 3,755 benchmark points with $y^+ \leq \min(150, 0.3Re_\tau)$, our transcription of their forms reproduces their reported picture: M16 gives 0.958 and PySR

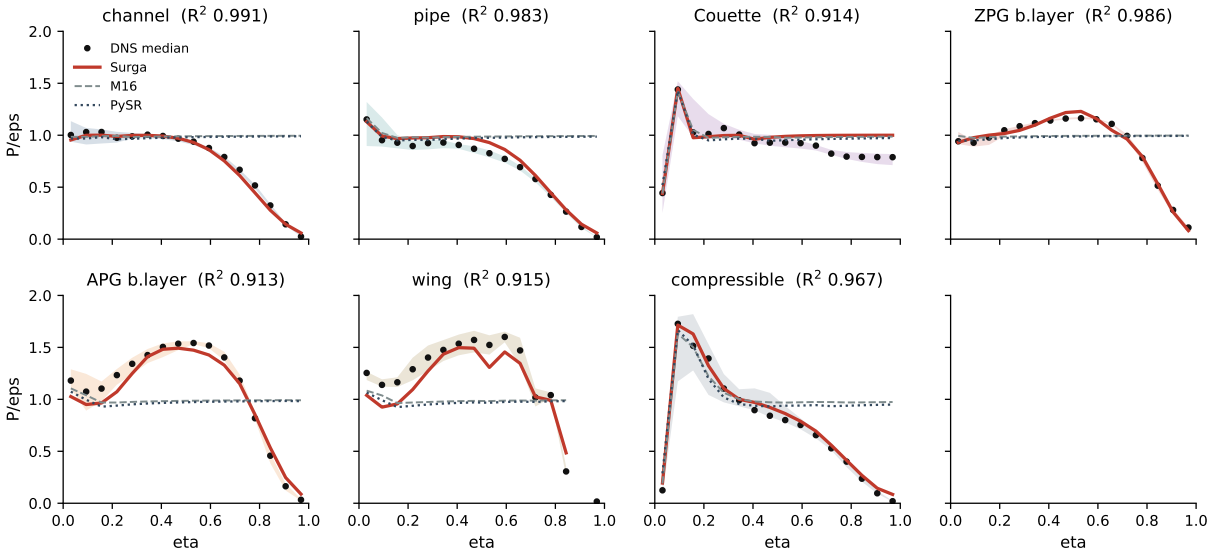


Figure 4: Median measured profile of each class (dots), interquartile band, and the three closures.

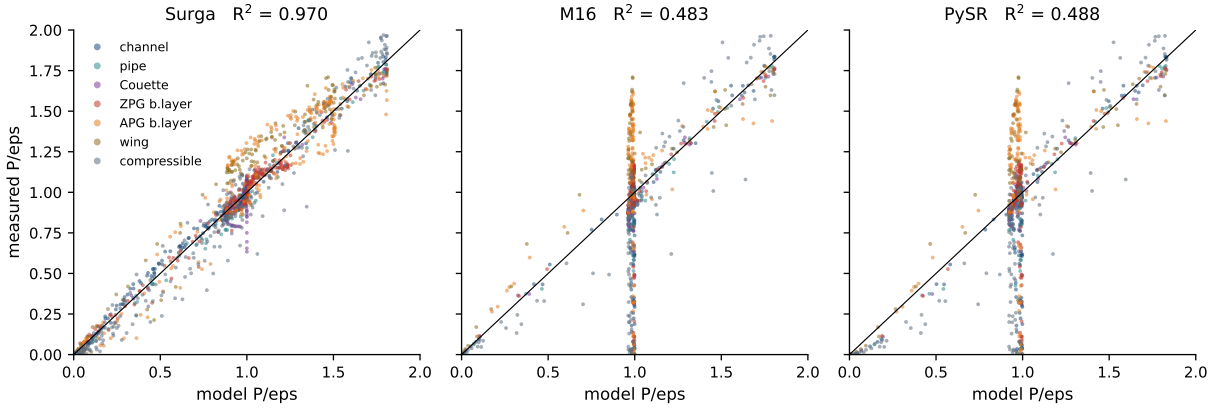


Figure 5: Model against measurement over all 52 profiles, thinned for legibility. Left: the present closure. Centre and right: the two published forms, which are functions of y^+ alone.

Table 5: Pooled R^2 and RMSE over all points of each class (mask described in the text).

flow class	points	Surga		M16		PySR	
		R^2	RMSE	R^2	RMSE	R^2	RMSE
plane channel	3214	0.989	0.038	0.234	0.322	0.242	0.321
pipe	1994	0.983	0.052	0.628	0.239	0.645	0.233
plane Couette	271	0.912	0.112	0.907	0.115	0.915	0.110
ZPG boundary layer	1458	0.986	0.044	0.509	0.258	0.510	0.258
APG boundary layer	1006	0.918	0.134	0.338	0.382	0.314	0.389
aerofoil wake	130	0.914	0.155	0.639	0.318	0.614	0.328
compressible channel	678	0.966	0.109	0.631	0.360	0.652	0.349
all	8751	0.970	0.072	0.483	0.302	0.488	0.300

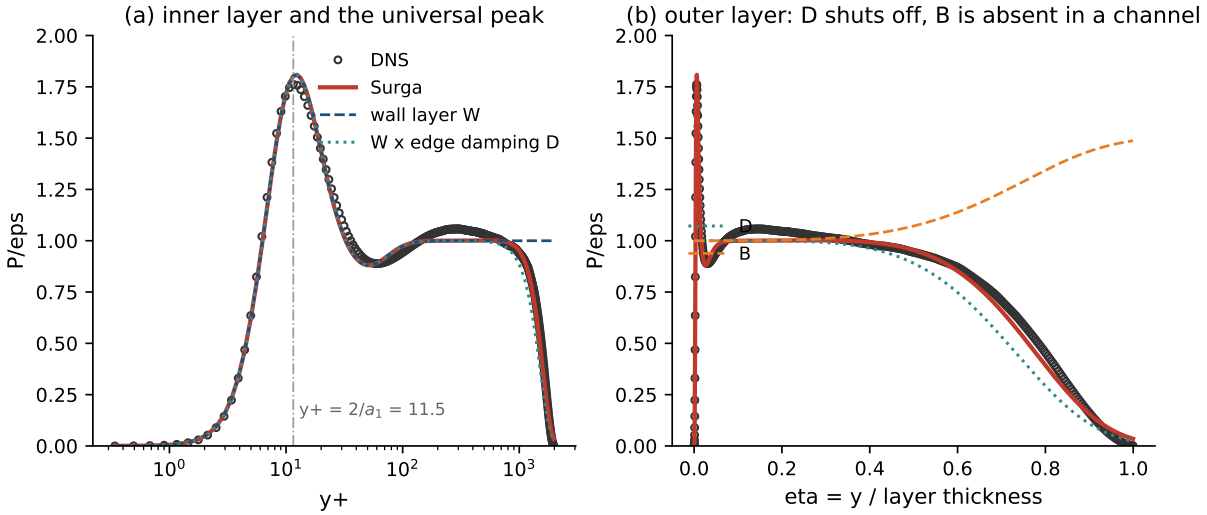


Figure 6: The three factors in a channel at $Re_\tau = 2000$. (a) On a logarithmic y^+ axis the wall factor reproduces the sublayer and the buffer peak; the dash-dot line marks $y^+ = 2/a_1$, where it saturates. (b) In the outer coordinate the edge factor takes the profile to zero; the surplus factor is identically unity because a channel has no surplus stress.

0.958, with about 0.99 on the channel and zero-pressure-gradient families. The present closure, evaluated on the same points, gives 0.973 with a lower RMSE (0.078 against 0.097). We take this as the useful part of the comparison: in the inner layer the three agree, as they should, because the inner layer is universal.

Over the whole layer. On the full mask the two published forms give 0.483 and 0.488, against 0.970 for the present fit. Restricting attention to the outer half of every profile, $\eta > 0.5$, isolates the reason (Table 7): in that half both published forms have a negative R^2 in six of the seven classes, which means they are then doing worse than the class mean. Their RMSE in that region is seven to thirteen times the present one. Their forms were not built for this region, and we read the table as a measurement of how far the inner-layer closure can be pushed rather than as a defect of that closure.

8 Reading the differences

The tables above are measurements; what follows is our reading of them, and it should be taken as such.

Table 6: Restricted to the intended inner domain, $y^+ \leq \min(150, 0.3Re_\tau)$.

flow class	points	R^2			RMSE		
		Surga	M16	PySR	Surga	M16	PySR
plane channel	895	0.994	0.989	0.992	0.035	0.046	0.041
pipe	1217	0.989	0.976	0.987	0.042	0.064	0.048
plane Couette	133	0.994	0.994	0.991	0.042	0.041	0.050
ZPG boundary layer	546	0.991	0.987	0.991	0.041	0.049	0.041
APG boundary layer	498	0.889	0.863	0.840	0.149	0.166	0.179
aerofoil wake	95	0.917	0.891	0.873	0.156	0.179	0.193
compressible channel	371	0.974	0.944	0.942	0.113	0.165	0.168
all	3755	0.973	0.958	0.958	0.078	0.097	0.097

Table 7: The outer half of every profile, $\eta > 0.5$. Median pred/obs is the median of the ratio of model to measurement.

flow class	points	R^2			RMSE			median pred/obs	
		Surga	M16	PySR	Surga	M16	PySR	M16	PySR
plane channel	1084	0.982	-2.062	-2.030	0.042	0.551	0.548	1.63	1.62
pipe	346	0.946	-2.984	-2.897	0.065	0.555	0.549	1.67	1.65
plane Couette	93	-2.023	-2.058	-1.664	0.179	0.180	0.168	1.16	1.14
ZPG boundary layer	464	0.981	-0.292	-0.283	0.053	0.441	0.439	1.05	1.05
APG boundary layer	306	0.958	-0.006	-0.011	0.113	0.555	0.556	0.84	0.83
aerofoil wake	17	0.909	-0.008	-0.014	0.167	0.557	0.559	0.83	0.82
compressible channel	217	0.877	-2.607	-2.348	0.109	0.591	0.569	1.96	1.92
all	2527	0.963	-0.774	-0.744	0.076	0.528	0.523	1.45	1.43

The large-y^+ limit is unity. Both published forms tend to $1 - a_4/y^+$ and never exceed unity. Unity is the statement that production balances dissipation pointwise. The inner layer is close to that; the outer layer is not, because energy produced near the wall is carried outward and, under a pressure gradient, the outer flow produces more than it dissipates. A form whose limit is unity can approach but not reach the measured 1.5–1.6 of the adverse-pressure-gradient and aerofoil cases. In our data the deficit is 16–17% over the outer half of those classes.

Nothing in the form locates the edge. Both expressions increase monotonically towards unity and never come back down, while the measured ratio falls to 0.03–0.1 at $\eta \to 1$ in the zero-pressure-gradient cases, where the two forms return 0.97–0.99. To know that a layer ends one has to know where it ends; $y^+ = 300$ can be 15% or 85% of the way to the edge depending on Re_τ, so y^+ cannot supply that information.

Nothing in the form sees the pressure gradient. An adverse pressure gradient thickens the wake and raises the mid-layer stress above the wall value, and both raise P/ε to 1.5–1.6. With one ceiling at unity the same function is too high where the data fall away (channel, zero-pressure-gradient) and too low where the data rise (adverse gradient, aerofoil): the median pred/obs in the outer half is 1.63 and 1.05 for the first group and 0.84 and 0.83 for the second.

Couette flow. In plane Couette flow the total stress is constant, so y^+ is only a rescaled wall distance, while $\langle u'v' \rangle$ must vanish at the mid-plane and the measured P/ε falls to 0.63–0.85 there. A function of y^+ cannot know the mid-plane exists and returns unity. The authors of [1] report that their formula fails out of domain on Couette flow; our scores are consistent with that, though we note the present closure also does poorly in that region (Section 9).

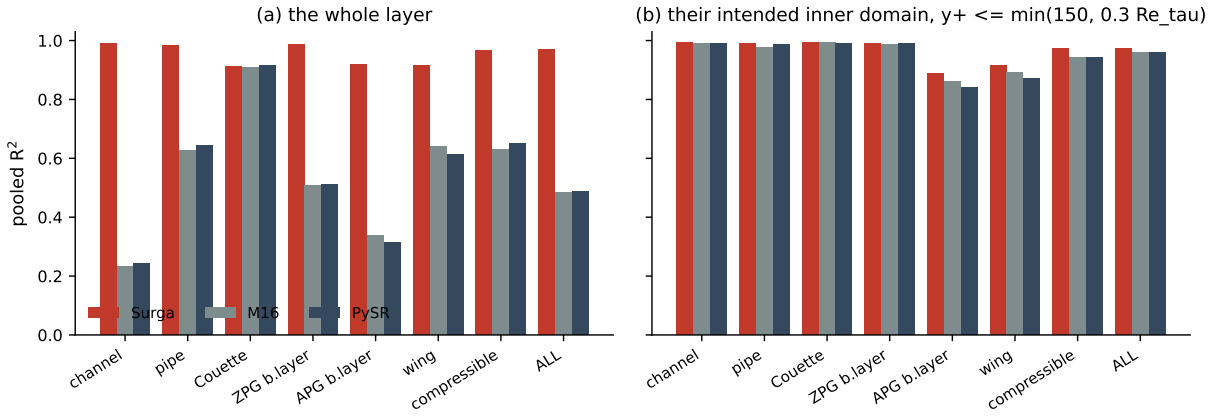


Figure 7: Pooled R^2 by class. (a) Over the whole layer. (b) Restricted to the intended inner domain, where the three closures are within a few hundredths of each other.

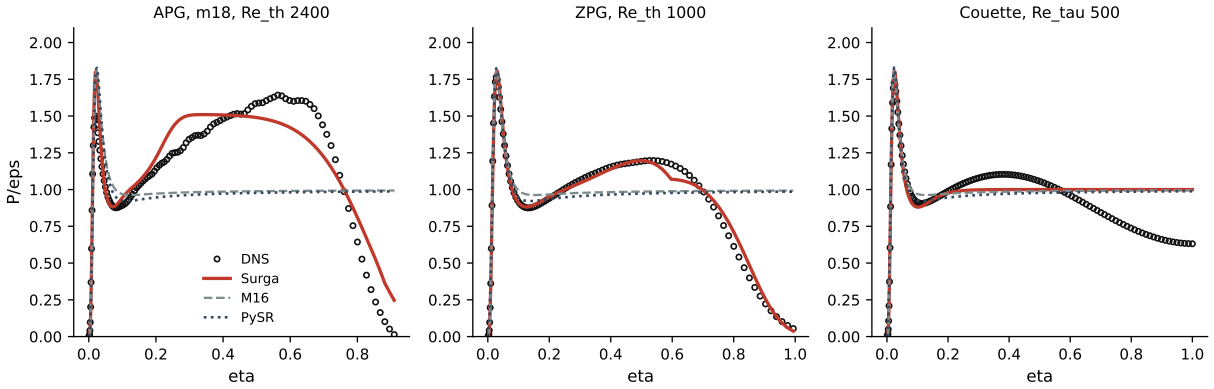


Figure 8: Three profiles, one per regime. Left: adverse-pressure-gradient layer at $Re_\theta = 2400$. Centre: zero-pressure-gradient layer. Right: plane Couette flow. Measurements are the open circles.

The wall exponent of Eq. (10). Expanding the linear tangent against a_4/y^+ gives $P/\varepsilon = (a_1/a_4)(y^+)^2 + O((y^+)^3)$, a quadratic wall law, which the no-slip constraint does not allow; the $\tanh^2$ numerator of Eq. (9) corrects it, as that paper states. With the printed constants the quadratic term is 2.1 times the measured value at $y^+ = 2$ and 0.21 times it at $y^+ = 20$: the error changes sign with distance, which is what a wrong exponent looks like.

One constant doing two jobs. In both forms a_4 fixes the near-wall coefficient, $c_w = a_1^2/a_4$, and also the size of the log-layer correction a_4/y^+ (their Eq. 26, where the value obtained from the log-layer transport coefficient is about 5.67). The two requirements pull in opposite directions, and the printed calibration lands at $c_w = 4.29 \times 10^{-3}$ against a DNS value of 4.96×10^{-3}, 13.6% low, as the authors report. In the present closure these two roles are played by different terms.

9 Limitations

The viscous sublayer is high. No single wall function satisfies both the exact cubic asymptote and the buffer peak. With a_1 pinned to the measured coefficient the residual bias is given in Table 8: the model is about 11% high at $y^+ < 1$ and about 1% low between $y^+ = 3$ and $y^+ = 5$. This is a bias on a quantity whose value there is of order 10^{-3}, so it contributes little to the reported errors, but it is real and it comes from the buffer bump reaching slightly into the sublayer. Letting a_1 float instead removes it at the cost of losing the exact asymptote.

Table 8: Median ratio of model to measurement in the near-wall bands.

wall distance	points	median model/measurement
$y^+ \in [0, 1)$	149	1.11
$y^+ \in [1, 3)$	262	1.05
$y^+ \in [3, 5)$	202	0.99
$y^+ \in [5, 8)$	237	0.94
$y^+ \in [8, 15)$	436	1.01
$y^+ \in [15, 30)$	635	1.00

Adverse-pressure-gradient layers. The class scores 0.918 in sample and 0.894 with the class held out, the weakest of the seven in the second test. The residual is concentrated at $\eta > 0.8$, where the model cuts the profile off slightly too late. The cut-off is written in η, while what actually ends production there is the vanishing of the mean shear, which the stress profile in the benchmark does not contain. Replacing η by a function of τ in the cut-off was tried and scored worse: the total stress in these layers falls steeply only in the last tenth of the layer.

Couette flow. This is the weakest class in sample (0.912) and has the worst profile in the leave-one-out test. The cut-off is switched by $(1 - \tau_e)$, which is identically zero in Couette flow, so the model predicts equilibrium across the outer half while the measurement falls to 0.63–0.85. In that outer half all three closures are in fact equally wrong (RMSE 0.167–0.179), so nothing in Section 7 should be read as the present fit winning this class; it does not. Closing the gap would need the mean shear, or the turbulent stress, at the outer boundary as an input, and adding a constant to reproduce three profiles seemed to us worse than stating the gap.

Compressible channels. This class scores 0.966. Part of the residual is a difference of definition rather than of model: in a Favre-averaged budget P and ε carry density weights, and the measured near-wall coefficient is 3.6×10^{-3} rather than 5.2×10^{-3}. The fit uses the incompressible coefficient and is correspondingly a few per cent high in the sublayer of those nine profiles.

10 Summary

The production-to-dissipation ratio of a wall-bounded flow can be written as a product of three statements. Near the wall it is a cube of the local wall distance, with a coefficient that appears to be common to all the flows examined here. Around $y^+ \approx 10$ it overshoots, because production is switched on by eddies that reach further than the dissipation can follow. Beyond that it becomes a statement about transport: where the flow carries only the stress the wall supplied the ratio returns to unity; where a pressure gradient or a moving boundary has injected extra stress it rises, apparently to a ceiling near 1.5; and where the layer ends it falls, abruptly at a free stream and gradually at a closed boundary. One set of nine numbers expresses that for channels, pipes, Couette flow, two kinds of boundary layer, an aerofoil wake and a compressible channel, with the weaknesses recorded in Section 9.

Data and code

The benchmark consists of 52 profiles assembled from published direct numerical simulations: channel data of Moser, Kim & Mansour [2], Lee & Moser [3] and Bernardini, Pirozzoli & Orlandi [4]; pipe data of El Khoury *et al.* [5]; plane Couette data of Lee & Moser [6]; zero-pressure-gradient boundary layers from the KTH database [7]; adverse-pressure-gradient boundary layers of Bobke *et al.* [8] together with the corresponding aerofoil cases; and compressible channel data

of Gerolymos & Vallet [9]. Each profile carries y^+, η, τ/τ_w and P/ε. Dissipation fields were cross-checked against the residual of the turbulent-kinetic-energy budget before use; two published fields that failed that check by factors of five to seventeen were corrected or dropped rather than fitted.

Code: `surga5.py` (the closure), `validate6.py` (fit and the two hold-out tests), `closures.py` (Eqs. (9) and (10) transcribed as printed).

A note on the status of these numbers

This document is an application note. The expression was fitted to data; the physical reading in Section 8 is an interpretation of that fit and not a derivation of it. The constants are empirical and nothing here has been validated outside the 52 profiles listed, for instance in a coupled RANS calculation. The numbers are reproducible from the accompanying code, and anyone relying on them should verify them independently; no warranty of accuracy is given for other flows, and no responsibility is accepted for consequences of such use.

A The closure as code

```
import numpy as np

C = dict(a1=0.17331, b1=0.22516, c1=0.10237, k=3.80596,
         r_conf=5.01887, r_free=8.05979, a=0.51082, q=3.73557,
         g=78.65106, h=1.89912)

def pep(yp, eta, tau, te, free_edge):
    eta = np.clip(eta, 0.0, 1.0)
    ys  = yp * np.sqrt(np.maximum(tau, 1e-6))
    T   = np.tanh(C["b1"]**2 * ys / 2.0)
    W   = np.tanh(C["a1"] * ys)**3 * (
          1.0 - T**5 * (1.0 - T**2) + C["c1"] * ys**2 * np.exp(-C["b1"] * ys))
    r   = np.where(free_edge, C["r_free"], C["r_conf"])
    D   = np.exp(-C["k"] * eta**r * (1.0 - te))
    X   = np.maximum(tau - (1.0 + (te - 1.0) * eta), 0.0)
    B   = 1.0 + C["a"] * np.tanh(
          eta**C["q"] * (C["g"] * X + C["h"] * (1.0 - te)))
    return W * D * B
```

References

[1] J. Jiang and B. Rosic, Combining symbolic regression and turbulence physics to develop a compact inner-layer production–dissipation closure, *Phys. Fluids* **38**, 095140 (2026).

[2] R. D. Moser, J. Kim and N. N. Mansour, Direct numerical simulation of turbulent channel flow up to $Re_\tau = 590$, *Phys. Fluids* **11**, 943–945 (1999).

[3] M. Lee and R. D. Moser, Direct numerical simulation of turbulent channel flow up to $Re_\tau \approx 5200$, *J. Fluid Mech.* **774**, 395–415 (2015).

[4] M. Bernardini, S. Pirozzoli and P. Orlandi, Velocity statistics in turbulent channel flow up to $Re_\tau = 4000$, *J. Fluid Mech.* **742**, 171–191 (2014).

[5] G. K. El Khoury, P. Schlatter, A. Noorani, P. F. Fischer, G. Brethouwer and A. V. Johansson, Direct numerical simulation of turbulent pipe flow at moderately high Reynolds numbers, *Flow Turbul. Combust.* **91**, 475–495 (2013).

[6] M. Lee and R. D. Moser, Extreme-scale motions in turbulence, *J. Fluid Mech.* **842**, 128–145 (2018).

[7] P. Schlatter and R. Örlü, Assessment of direct numerical simulation data of turbulent boundary layers, *J. Fluid Mech.* **659**, 116–126 (2010).

[8] A. Bobke, R. Vinuesa, R. Örlü and P. Schlatter, History effects and near equilibrium in adverse-pressure-gradient turbulent boundary layers, *J. Fluid Mech.* **820**, 667–692 (2017).

[9] G. A. Gerolymos and I. Vallet, Pressure, density, and temperature fluctuations in turbulent channel flows up to $M = 2$, *J. Fluid Mech.* **757**, 701–746 (2014).

[10] I. Marusic and J. P. Monty, Attached eddy model of wall turbulence, *Annu. Rev. Fluid Mech.* **51**, 49–74 (2019).